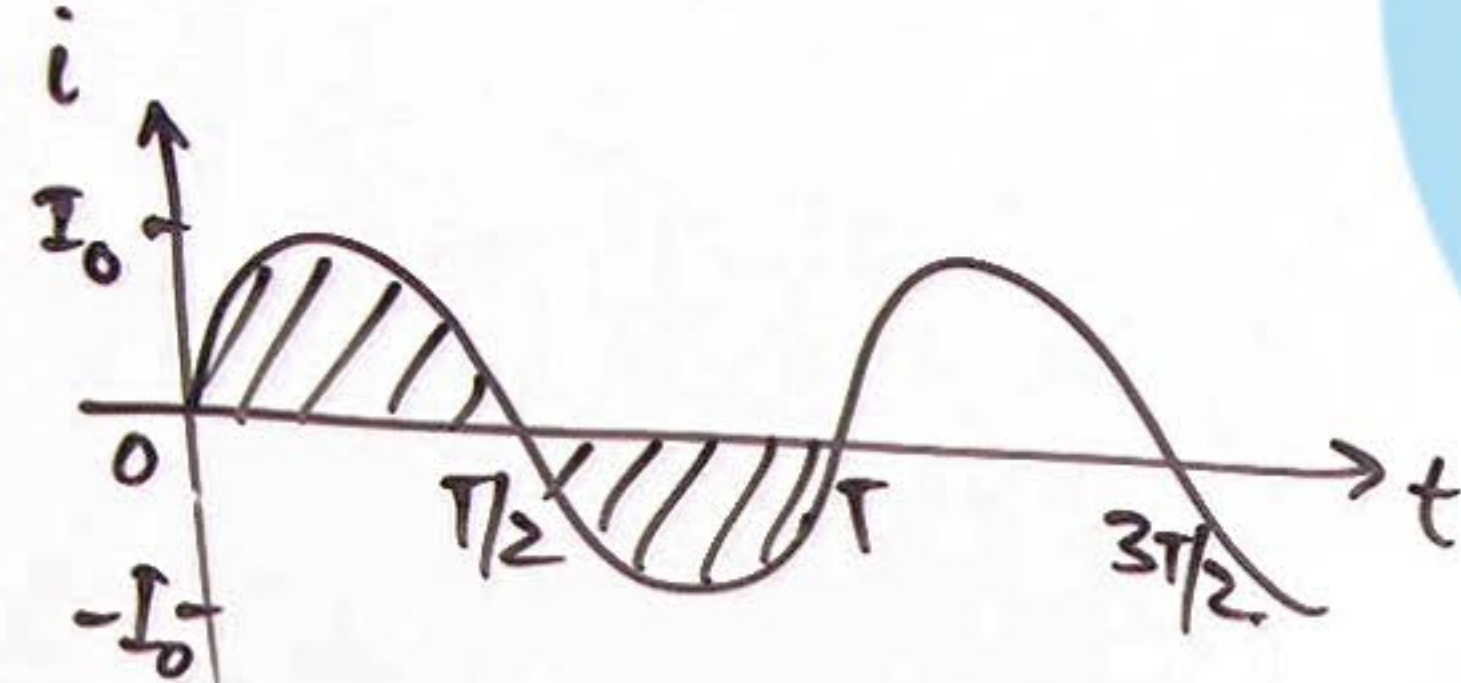
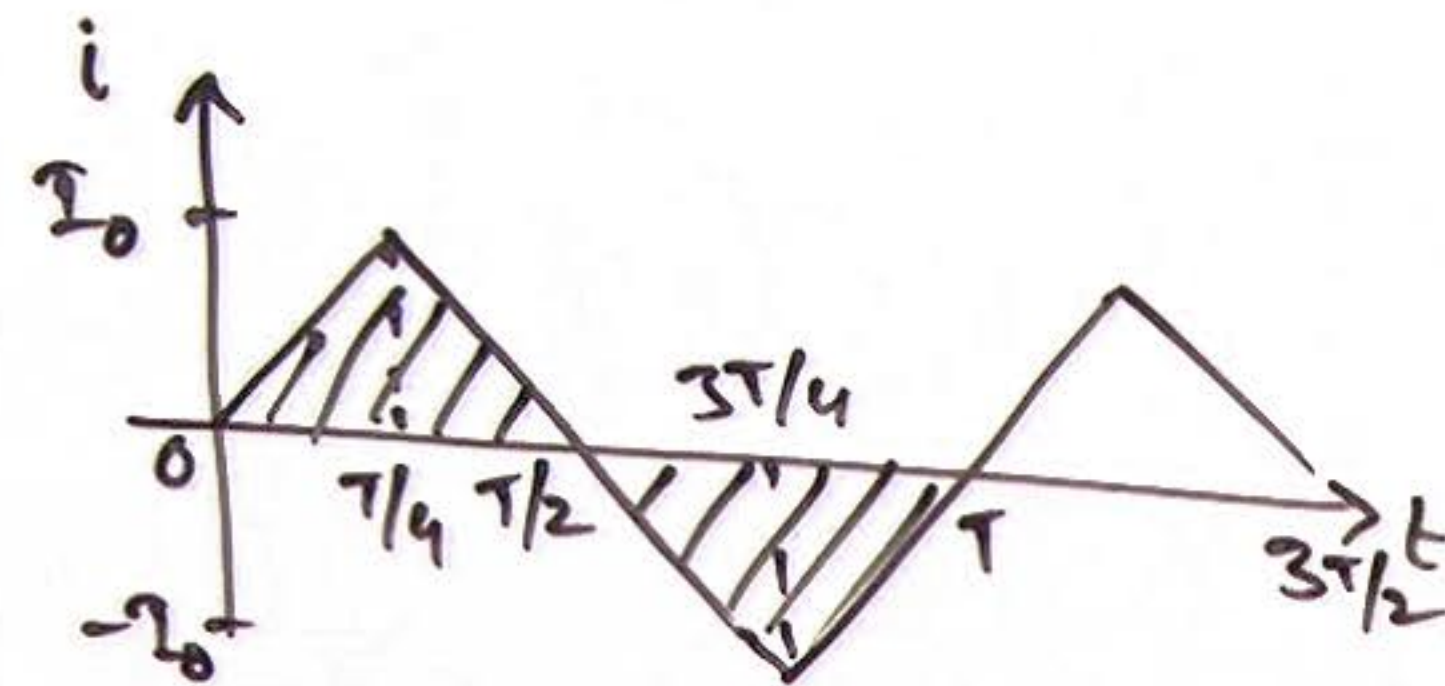
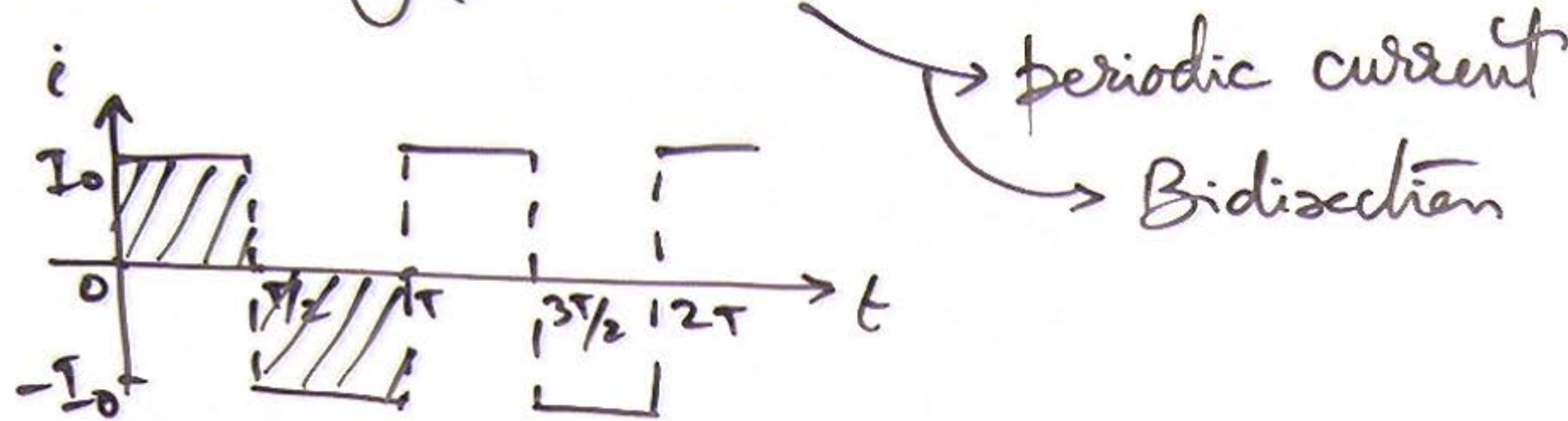


① Alternating Current :



② Mean & RMS values of AC :

$$I_{avg} = \frac{\Delta q}{\Delta t} = \frac{\text{Area under } i-t \text{ curve}}{\Delta t}$$

$$I_{avg} = \frac{1}{\Delta t} \int_0^{\Delta t} i(t) dt$$

$$I_{rms} = \sqrt{\frac{1}{\Delta t} \int_0^{\Delta t} i^2(t) dt}$$

$$I_{avg} = \frac{1}{t_2 - t_1} \int_{t_1}^{t_2} i(t) dt$$

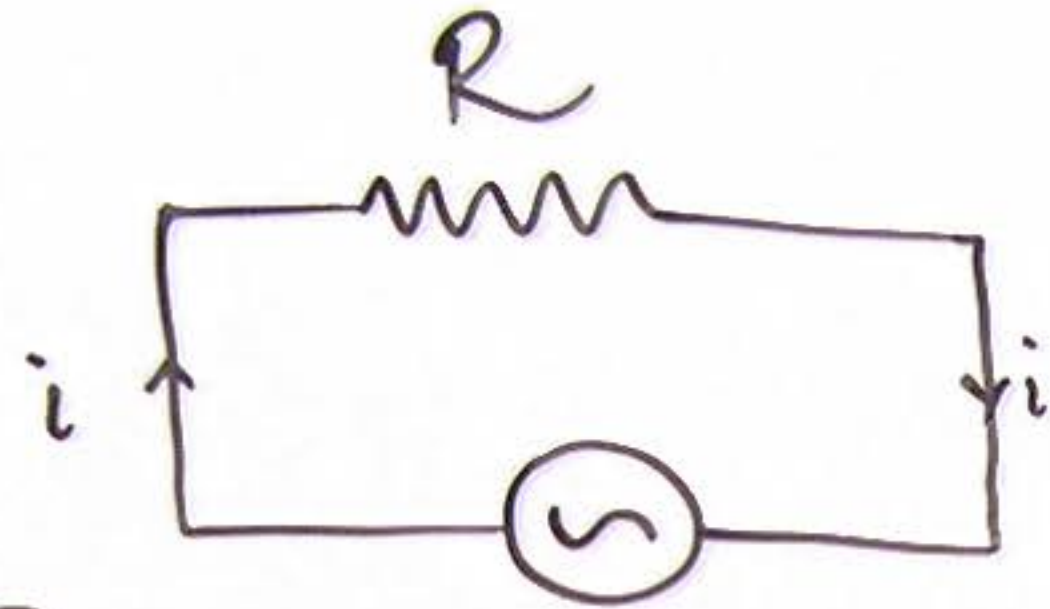
$$I_{rms} = \sqrt{\frac{1}{t_2 - t_1} \int_{t_1}^{t_2} i^2(t) dt}$$

for a Symmetric bidirectional AC

area & shape of +ve & -ve half cycle should be same. $I_{avg} = 0$

mainly use for power analysis & AC ckts.

② Circuit Components across AC emf:

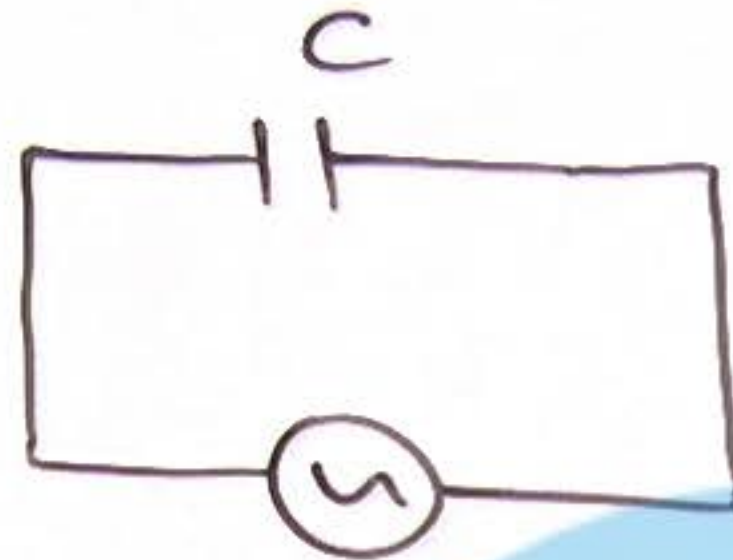


$$e = e_0 \sin \omega t$$

$$i = i_0 \sin \omega t$$

$$i_0 = \frac{e_0}{R}$$

Resistance
in Ohms

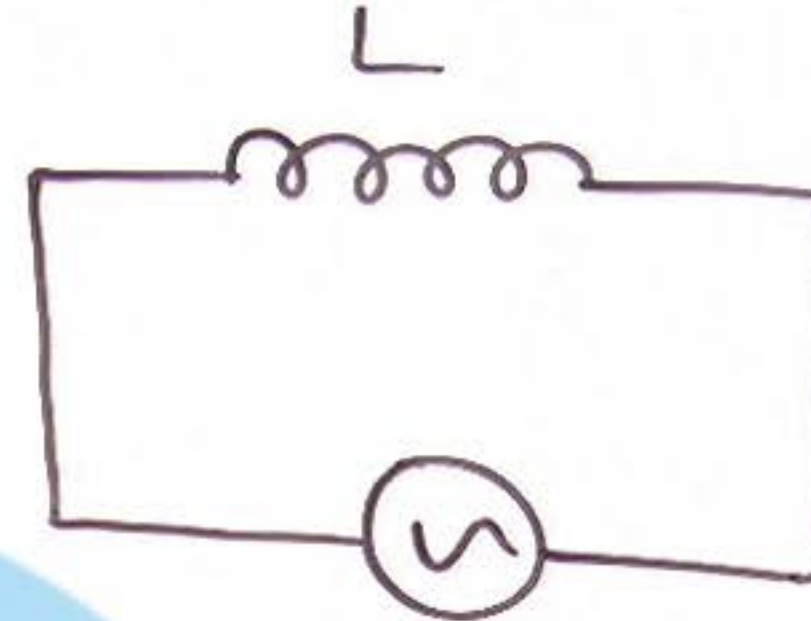


$$e = e_0 \sin \omega t$$

$$i = i_0 \sin(\omega t + \pi/2)$$

$$i_0 = e_0 C \omega = \frac{e_0}{(1/C\omega)}$$

$$X_C = \frac{1}{C\omega} \Omega \text{ [Capacitive Reactance]}$$



$$e = e_0 \sin \omega t$$

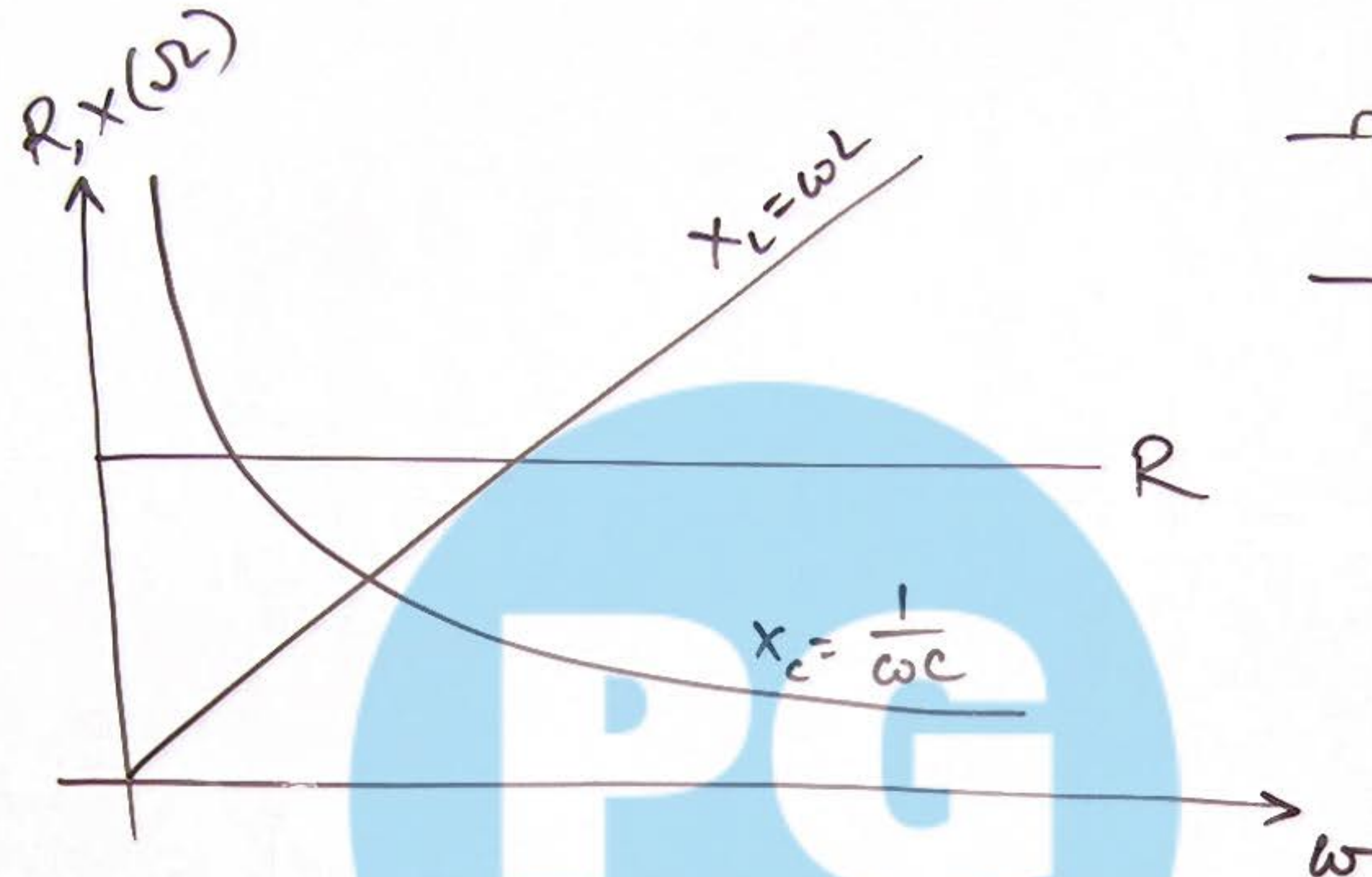
$$i = i_0 \sin(\omega t - \pi/2)$$

$$i_0 = \frac{e_0}{L\omega}$$

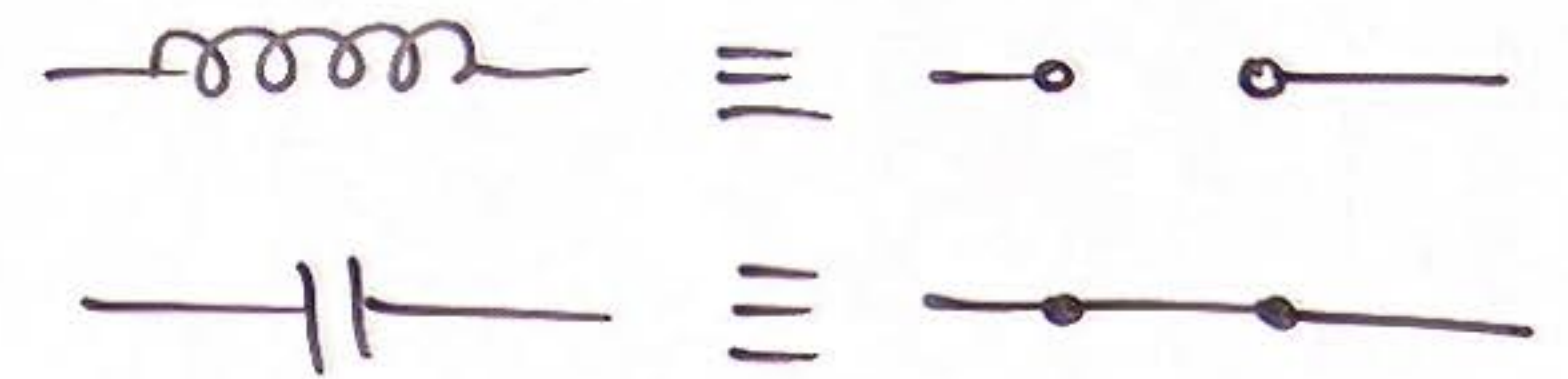
$$X_L = L\omega \Omega \text{ [Inductive Reactance]}$$

Same
phase

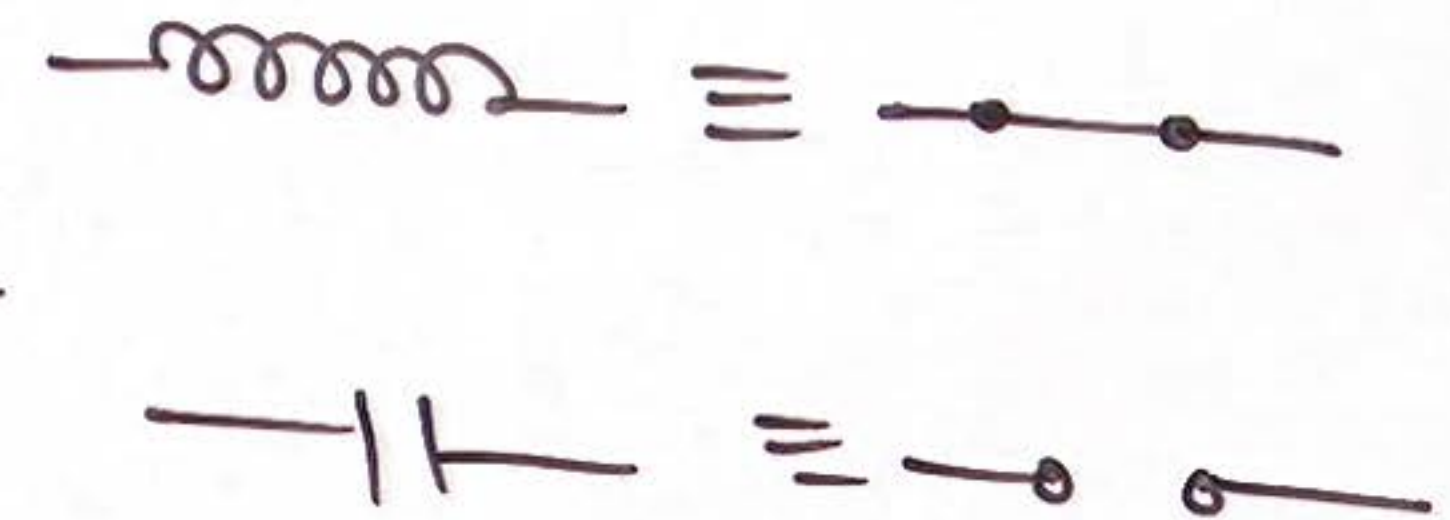
③ Comparison of AC circuit elements:



V. High freq.



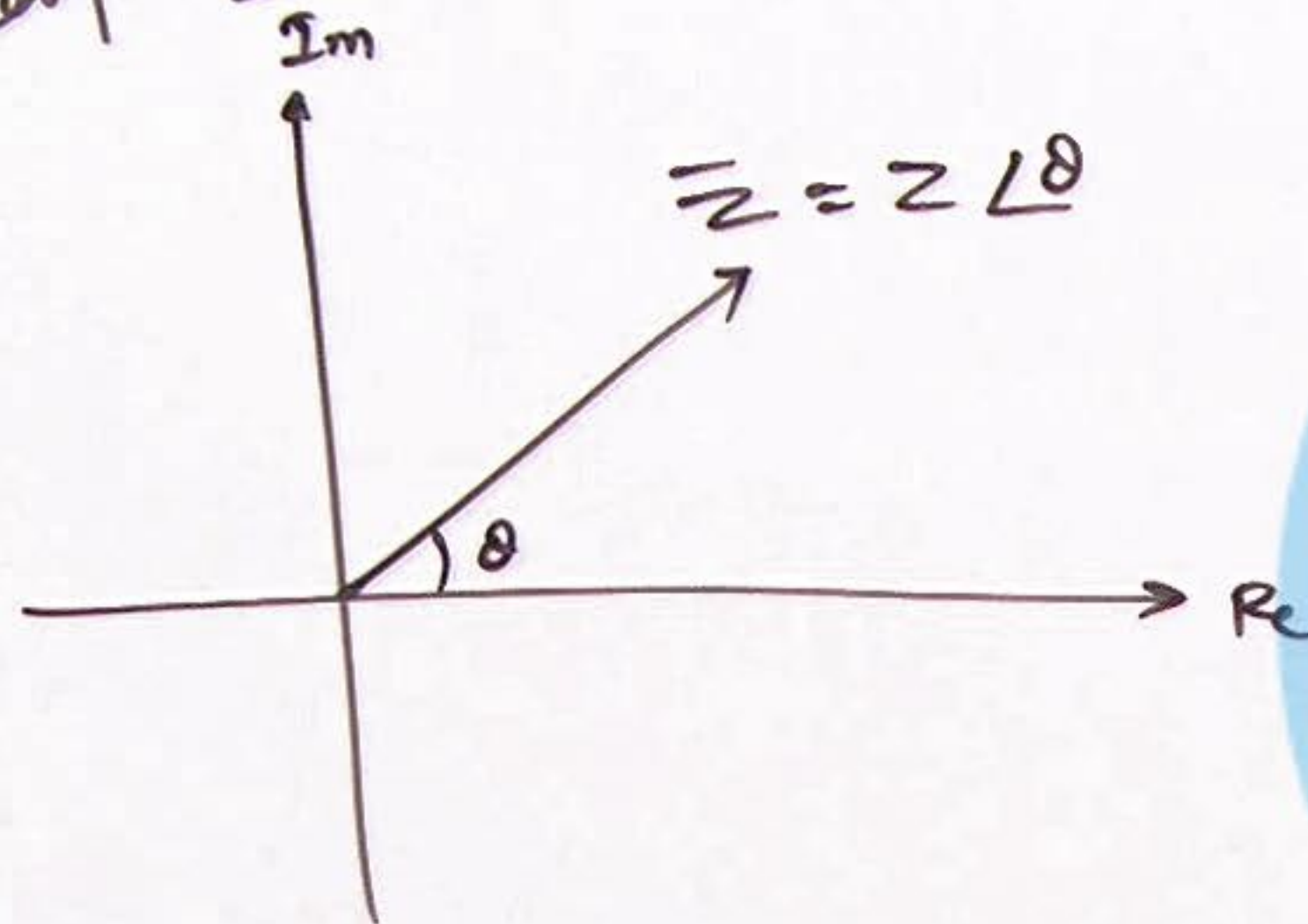
V. low freq. (almost DC)



④ Phasor Analysis :

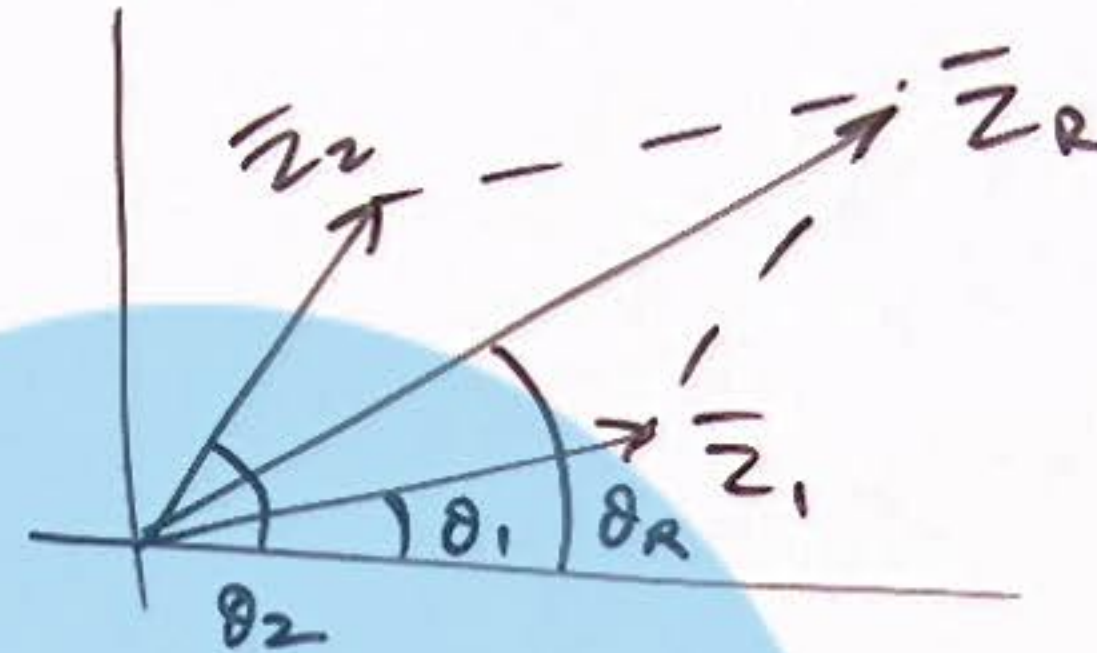
Similar to Vector

Also similar to
Complex Nos.



Addition of phasors

$$\bar{Z}_R = \bar{Z}_1 + \bar{Z}_2$$



Multiplication/Division of phasors.

$$\bar{Z}_1 = Z_1 \angle \theta_1 \quad \& \quad \bar{Z}_2 = Z_2 \angle \theta_2$$

$$\bar{Z}_m = \bar{Z}_1 \cdot \bar{Z}_2 = Z_1 Z_2 \angle \theta_1 + \theta_2$$

$$\bar{Z}_d = \frac{\bar{Z}_1}{\bar{Z}_2} = \frac{Z_1}{Z_2} \angle \theta_1 - \theta_2$$

⑤ Phasors in AC ckt :

$$\bar{e} = e_0 \angle 0^\circ (\text{Ref})$$

$$\bar{i} = i_0 \angle \phi$$

$$\bar{R} = R \angle 0^\circ$$

$$\bar{X}_C = \frac{1}{\omega C} \angle -\pi/2$$

$$\bar{X}_L = \omega L \angle +\pi/2$$

⑥ Impedance of AC ckt:

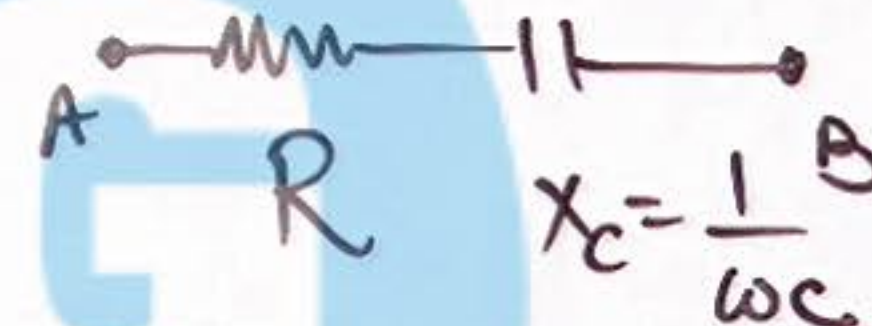
Total Ohmic
opposition of a ckt
across AC entire.

Ckt Impedance

$$\bar{Z} = \bar{R} + \bar{X} \quad \Omega$$

Resistance $\rightarrow R \quad \Omega$

Cap/Inductors $\rightarrow X_C = \frac{1}{\omega C} \quad \Omega$
 $X_L = \omega L \quad \Omega$

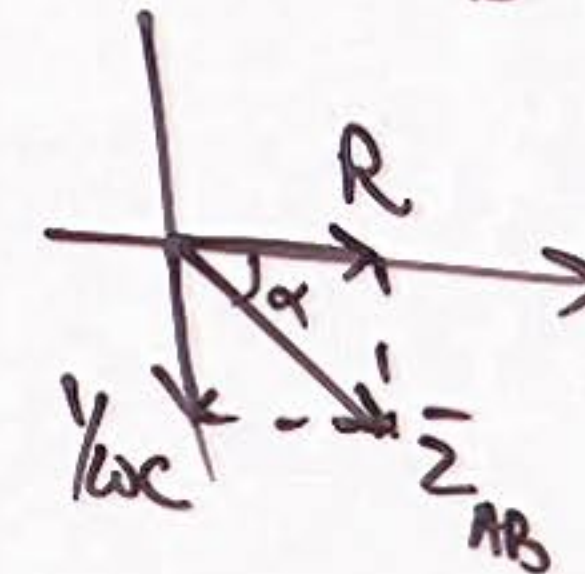


$$\bar{R} = R \angle 0^\circ$$

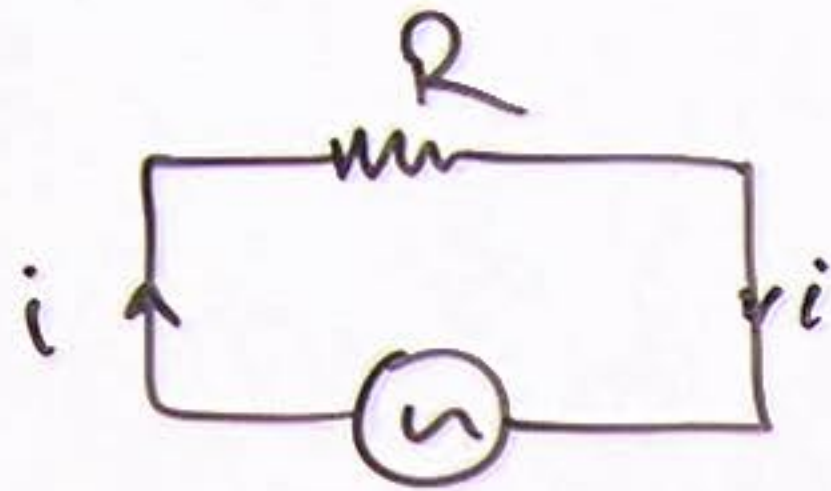
$$\bar{X}_C = \frac{1}{\omega C} \angle -\pi/2$$

$$\bar{Z}_{AB} = \bar{R} + \bar{X}_C$$

$$\bar{Z}_{AB} = \sqrt{R^2 + \frac{1}{\omega^2 C^2}} \angle -\tan^{-1} \frac{1}{\omega RC}$$



⑦ Phasor Analysis of R, L & C across AC emf:



$$e = e_0 \sin \omega t$$

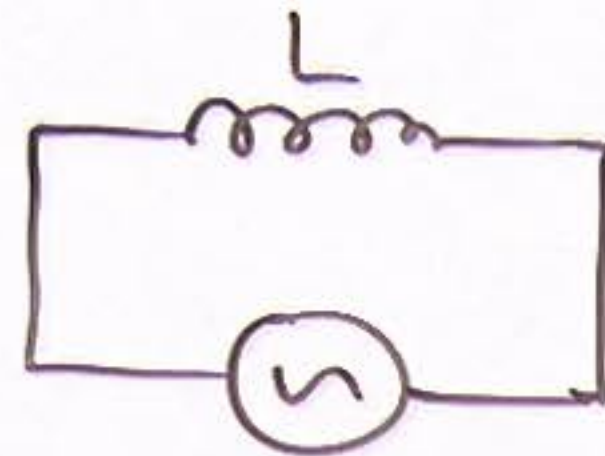
$$\bar{e} = e_0 \angle 0^\circ$$

By Ohm's Law -

$$\bar{i} = \frac{\bar{e}}{Z} = \frac{e_0 \angle 0^\circ}{R \angle 0^\circ}$$

$$\bar{i} = \frac{e_0}{R} \angle 0^\circ$$

$$i = \frac{e_0}{R} \sin \omega t$$



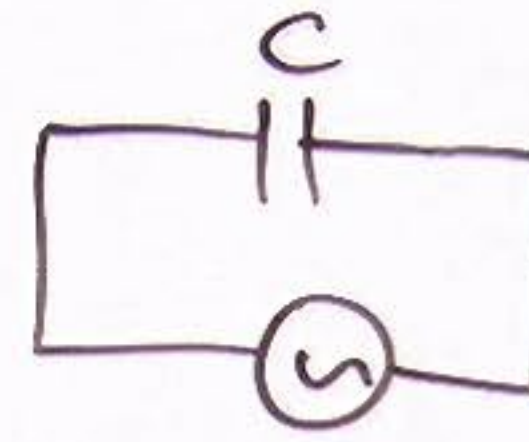
$$e = e_0 \sin \omega t$$

$$\bar{e} = e_0 \angle 0^\circ$$

$$\bar{i} = \frac{\bar{e}}{Z} = \frac{e_0 \angle 0^\circ}{\omega L \angle +\pi/2}$$

$$\bar{i} = i_0 \angle -\pi/2$$

$$i = \frac{e_0}{\omega L} \sin(\omega t - \pi/2)$$



$$e = e_0 \sin \omega t$$

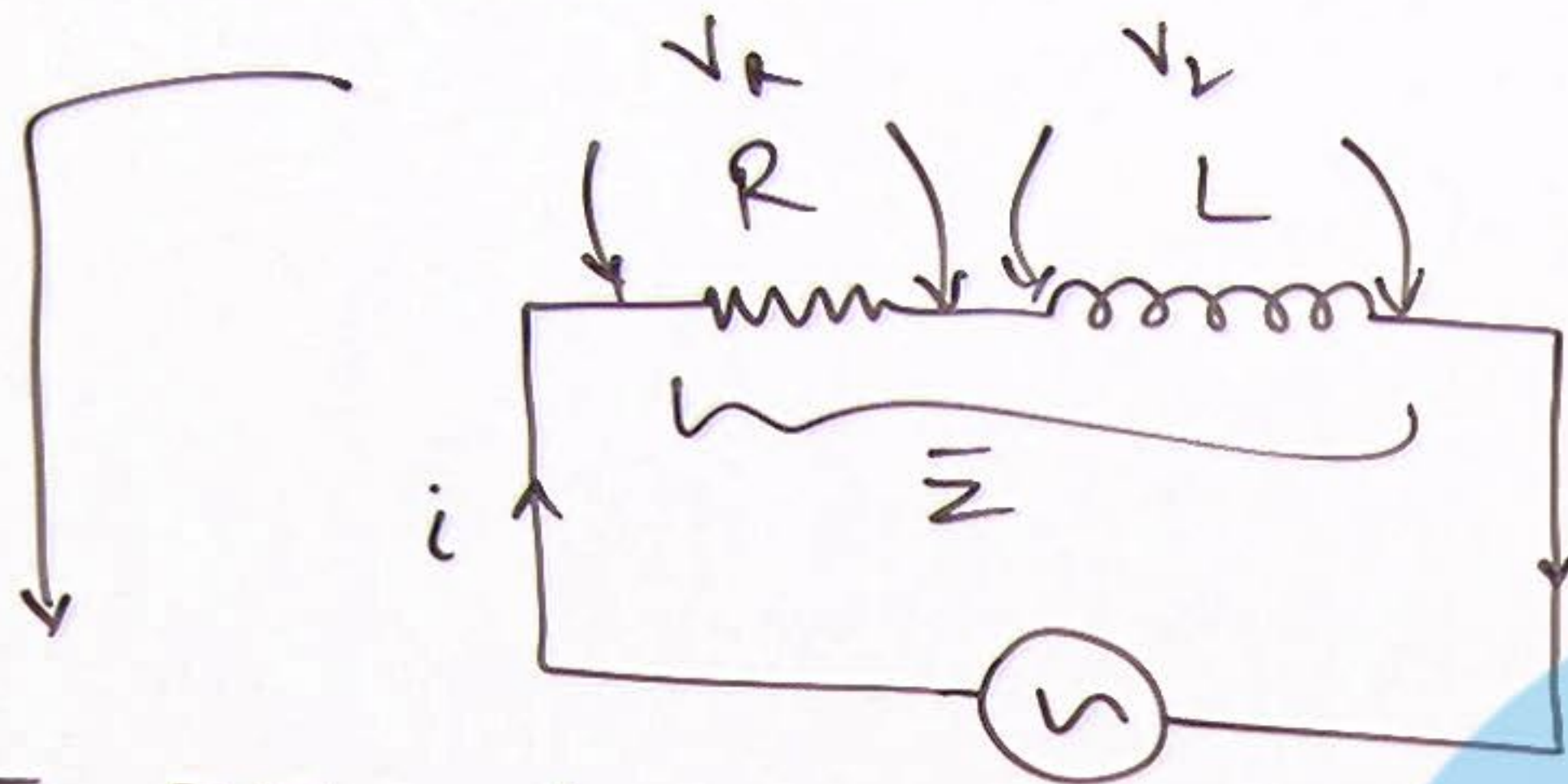
$$\bar{e} = e_0 \angle 0^\circ$$

$$\bar{i} = \frac{\bar{e}}{Z_c} = \frac{e_0 \angle 0^\circ}{\frac{1}{\omega C} \angle -\pi/2}$$

$$\bar{i} = e_0 \omega C \angle +\pi/2$$

$$i = i_0 \sin(\omega t + \pi/2)$$

⑧ RL Circuit across AC enf:



$$\bar{V}_R = \bar{i} \bar{R} = \checkmark$$

$$e = e_0 \sin \omega t$$

$$\bar{V}_L = \bar{i} \bar{X}_L$$

$$\bar{e} = e_0 \angle 0^\circ$$

$$\bar{V}_L = i_0 \omega L \left[90^\circ - \tan^{-1} \frac{\omega L}{R} \right] \checkmark$$

$$\bar{Z} = \bar{R} + \bar{X}_L$$

$$\bar{Z} = R \angle 0^\circ + \omega L \angle +\pi/2$$

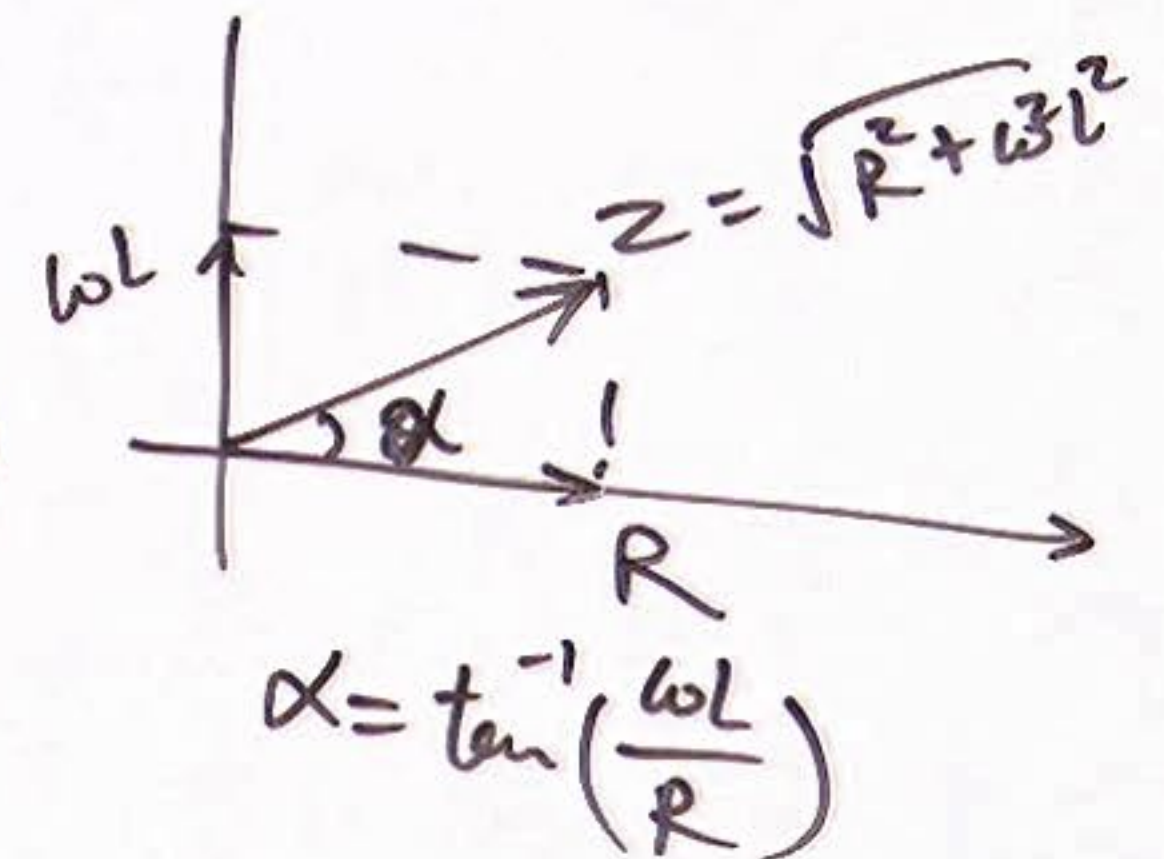
$$\bar{Z} = \sqrt{R^2 + \omega^2 L^2} \tan^{-1} \left(\frac{\omega L}{R} \right)$$

By Ohm's law

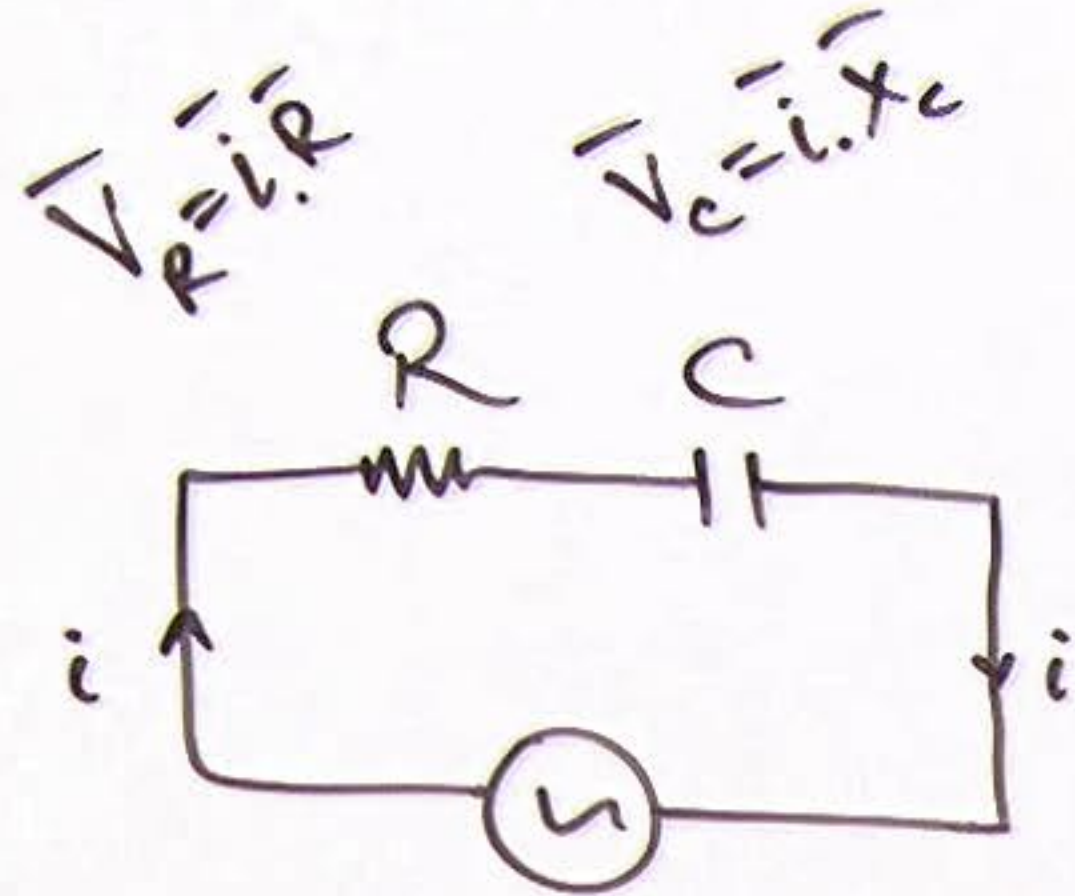
$$\bar{i} = \frac{\bar{e}}{\bar{Z}} = \frac{e_0 \angle 0^\circ}{\sqrt{R^2 + \omega^2 L^2} \angle \tan^{-1} \frac{\omega L}{R}}$$

$$\bar{i} = \left(\frac{e_0}{\sqrt{R^2 + \omega^2 L^2}} \right) \angle -\tan^{-1} \frac{\omega L}{R}$$

$$\Rightarrow i = \underbrace{i_0}_{\frac{e_0}{\sqrt{R^2 + \omega^2 L^2}}} \sin(\omega t + \underbrace{\alpha}_{-\tan^{-1} \frac{\omega L}{R}})$$



⑨ RC & LC circuit across AC emf:

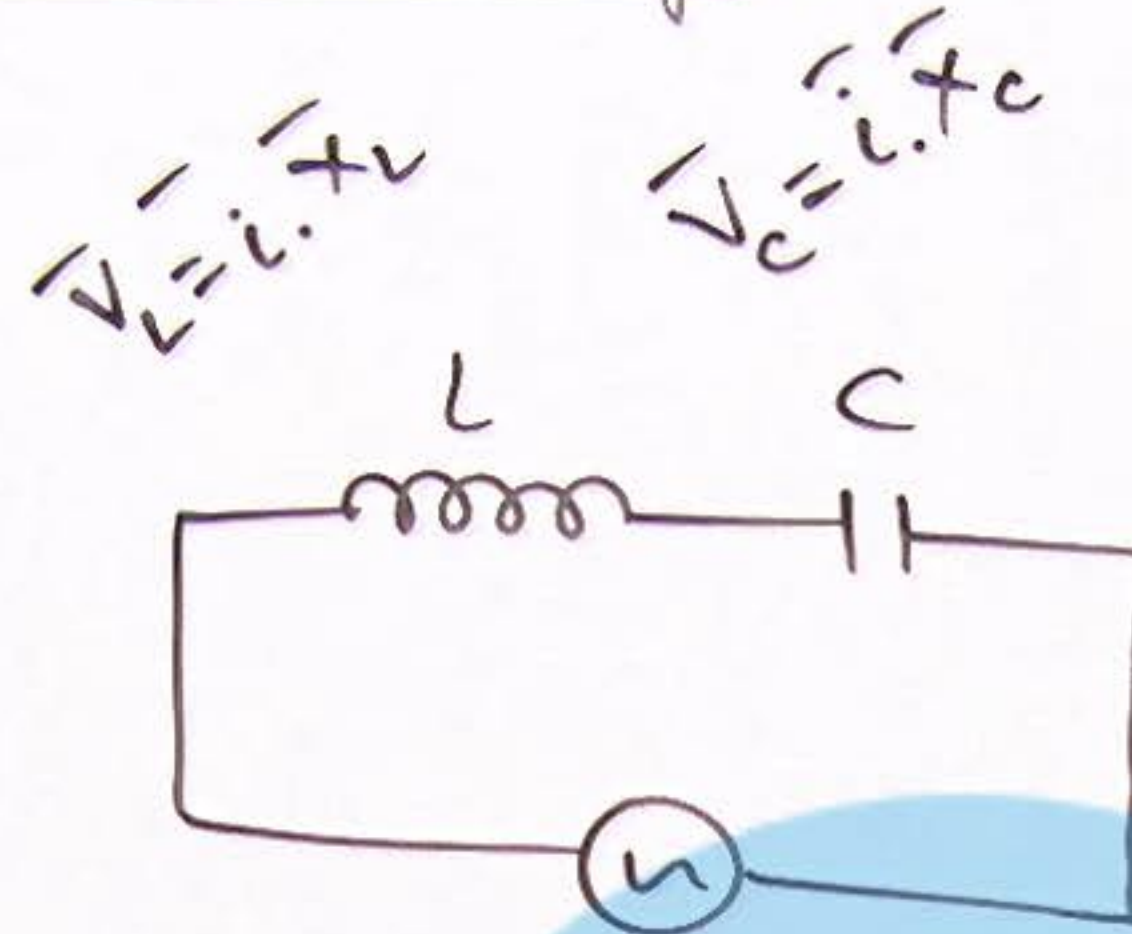


$$e = e_0 \sin \omega t$$

$$\bar{e} = e_0 \angle 0^\circ$$

$$\bar{i} = i_0 \angle \alpha \rightarrow i_0 \sin(\omega t + \alpha)$$

$$i_0 = \frac{e_0}{\sqrt{R^2 + \frac{1}{\omega^2 C^2}}} \quad \& \quad \alpha = \tan^{-1} \left(\frac{1}{\omega R C} \right)$$



$$e = e_0 \sin \omega t$$

$$\bar{e} = e_0 \angle 0^\circ$$

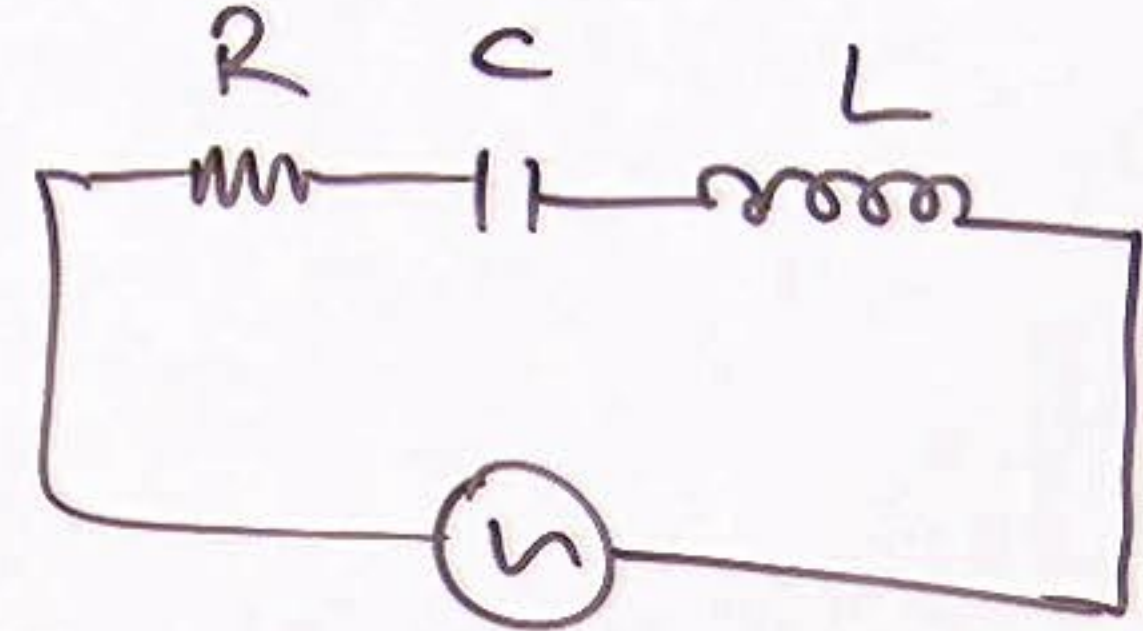
$$\bar{i} = i_0 \angle \alpha$$

$$i_0 = \frac{e_0}{|\omega L - \frac{1}{\omega C}|} \quad \& \quad \alpha = \tan^{-1} \left(\frac{1}{\omega C} - \omega L \right)$$

⑩ Series RLC ckt across AC emf:

$\alpha = \tan^{-1} \left(\frac{\omega L - \frac{1}{\omega C}}{R} \right)$

$Z = \sqrt{R^2 + (\omega L - \frac{1}{\omega C})^2}$

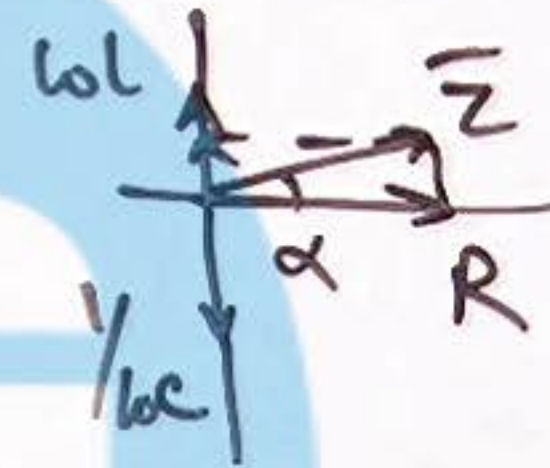


$$e = e_0 \sin \omega t$$

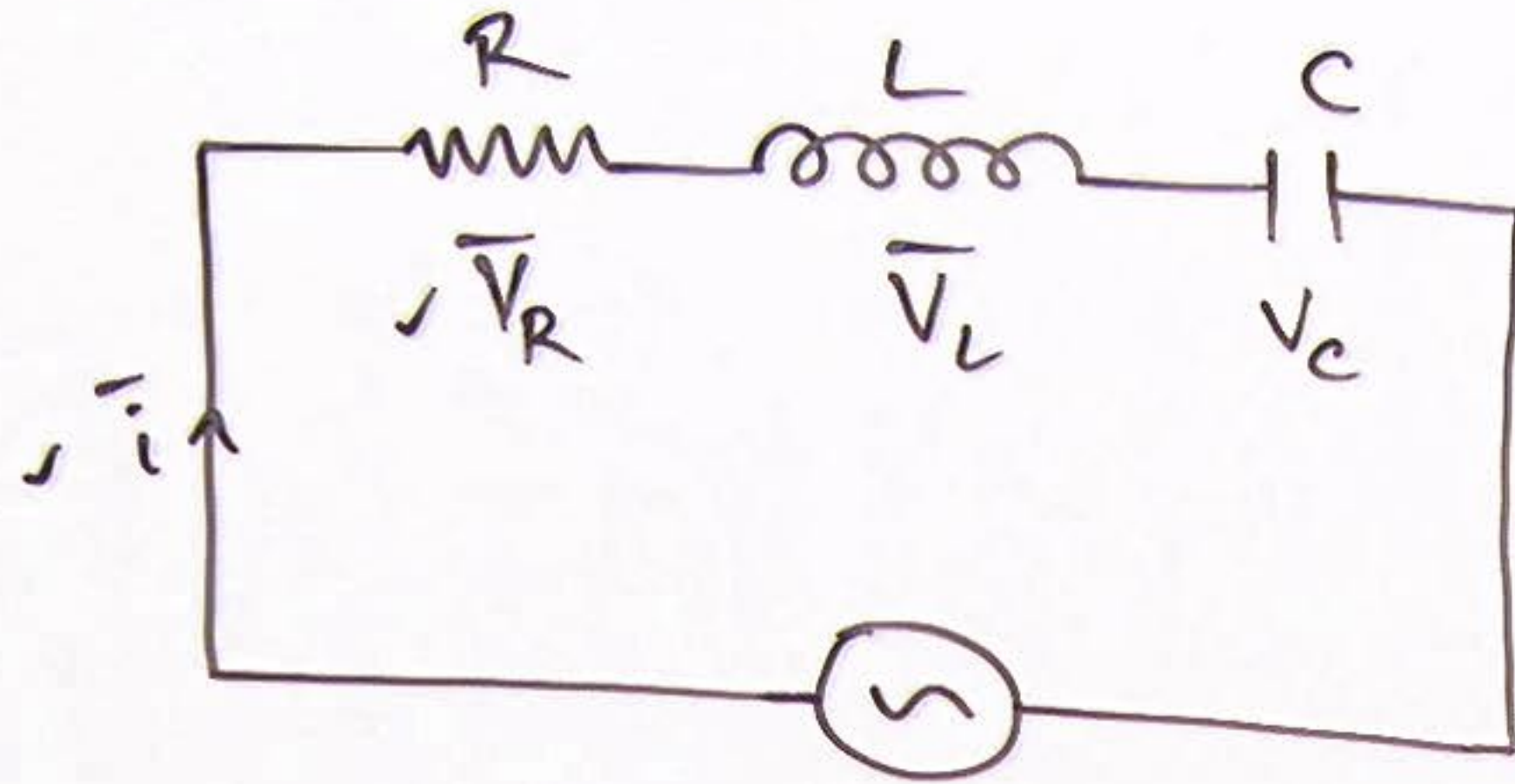
$$\bar{e} = e_0 \angle 0^\circ$$

$$i = i_0 \angle -\alpha$$

$$i_0 = \frac{e_0}{\sqrt{R^2 + (\omega L - \frac{1}{\omega C})^2}} \quad \checkmark$$



10 Phase Relations in Series R-L-C Circuit:



$$\bar{V}_R = \bar{i} \cdot \bar{R}$$

$$\bar{V}_L = \bar{i} \cdot \bar{X}_L$$

$$\bar{V}_C = \bar{i} \cdot \bar{X}_C$$

$$e = e_0 \sin \omega t$$

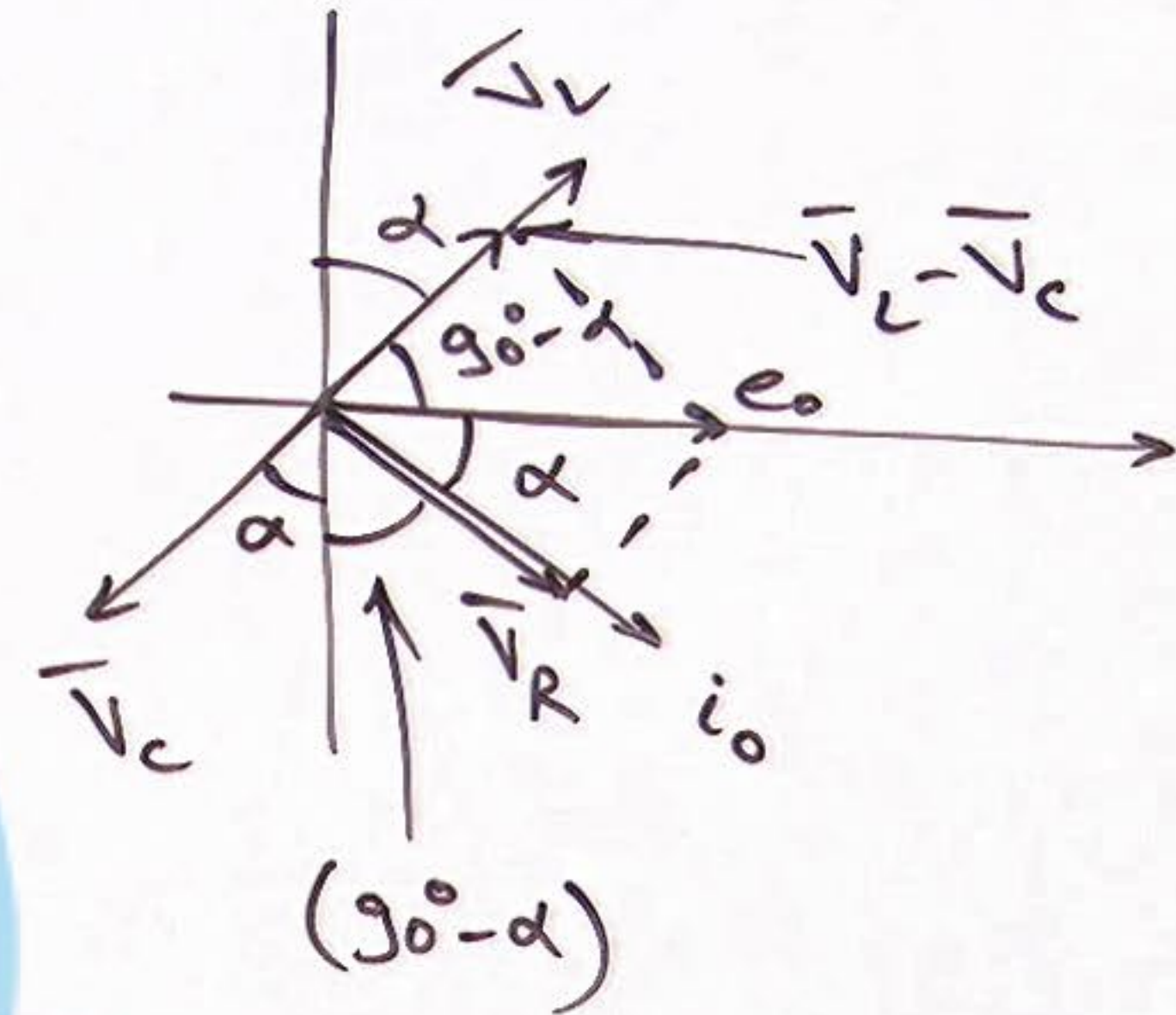
$$\bar{e} = e_0 \angle 0^\circ$$

$$i = i_0 \tan^{-1} \left(\frac{\omega L - 1/\omega C}{R} \right) \rightarrow \alpha$$

$$i_0 = \frac{e_0}{\sqrt{R^2 + (\omega L - 1/\omega C)^2}}$$

$$\bar{e} = \bar{V}_R + \bar{V}_L + \bar{V}_C$$

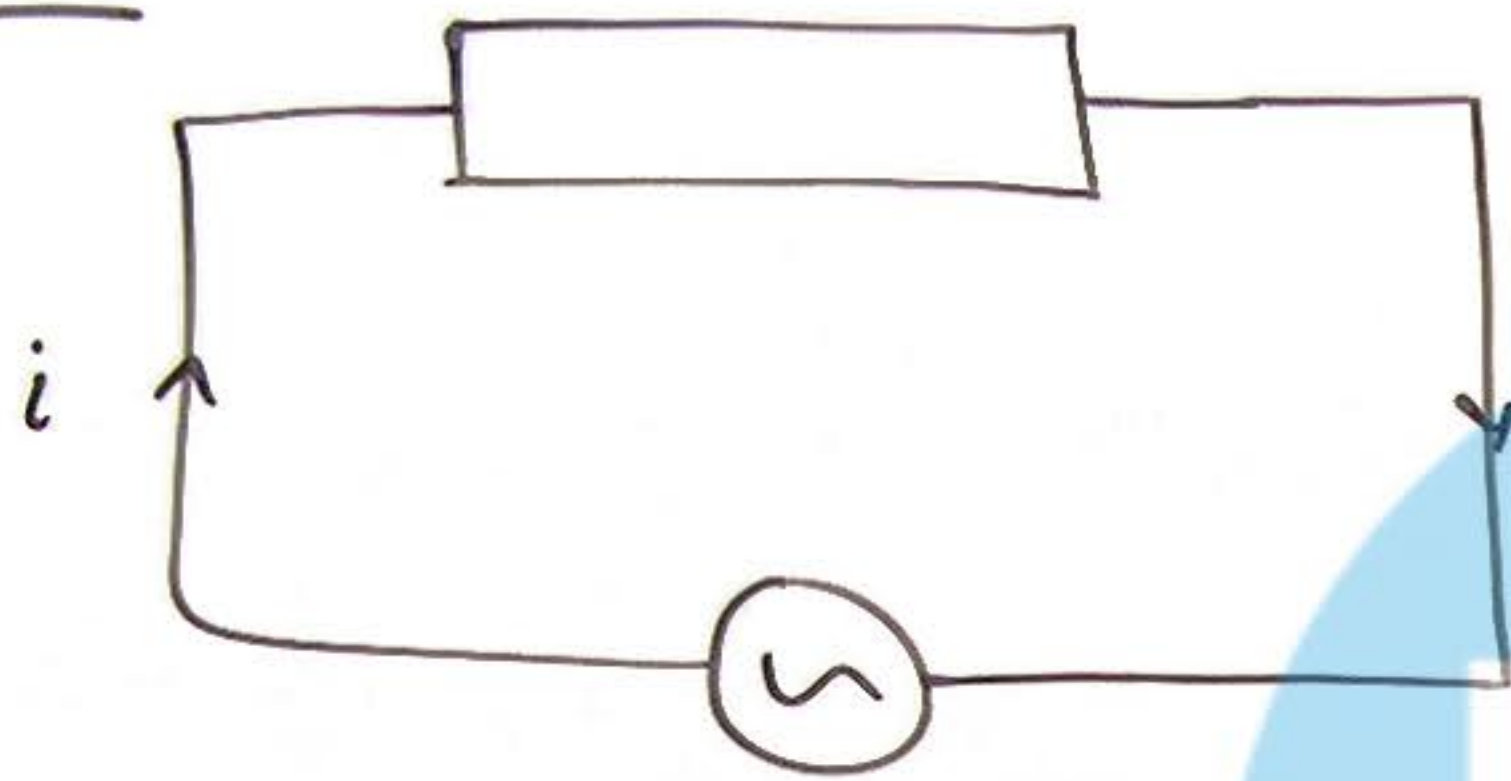
$$e_0 = \sqrt{V_R^2 + (V_L - V_C)^2}$$



(11) Power in AC Circuits :

If $Z = R$ (pure resistance)

$$\Rightarrow \underline{P_{avg}} = e_{rms} \cdot i_{rms} \quad \underline{Z = Z_0 \angle \phi}$$



$$i_0 = \frac{e_0}{Z_0} \angle -\phi$$

$$e = e_0 \sin \omega t$$

$$\bar{e} = e_0 \angle 0^\circ$$

$$i = i_0 \sin(\omega t - \phi)$$

Instant Power

$$P = ei = e_0 i_0 \sin \omega t \cdot \sin(\omega t - \phi)$$

$$P_{avg} = e_{rms} \cdot i_{rms} \cdot \cos \phi$$

$\cos \phi$
↑

power factor

$$e_{rms} = \frac{e_0}{\sqrt{2}} \quad \& \quad i_{rms} = \frac{i_0}{\sqrt{2}}$$

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⑫ Power in AC ckt for different Impedance:

$$P_{avg} = E_{rms} \cdot i_{rms} \cdot \cos \phi$$

$$\bar{Z} = \bar{R} + \bar{X}$$

Impedance $\rightarrow$ purely Resistive

$$\text{if } Z = R \Rightarrow \cos \phi = 1$$

$$\Rightarrow P_{avg} = E_{rms} i_{rms} [\text{Max}]$$

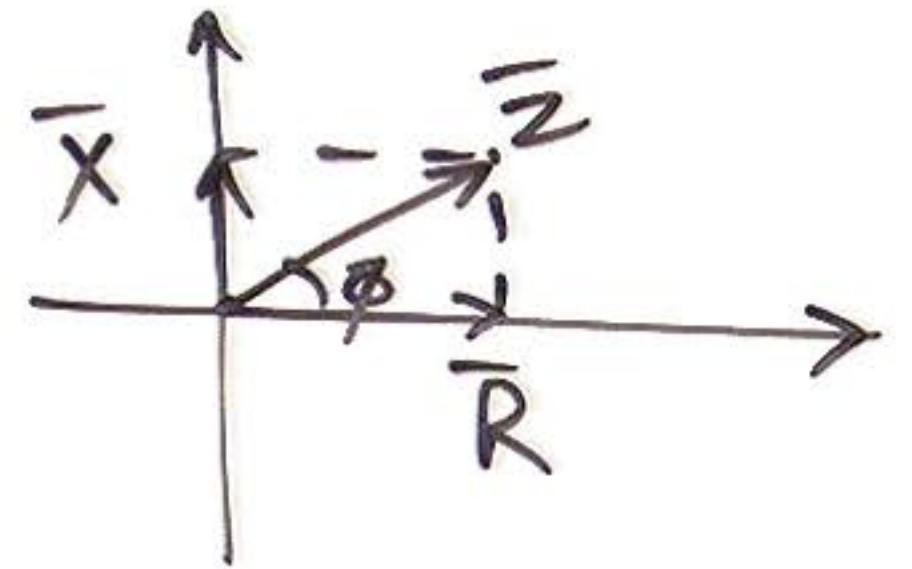
$$\text{if } Z = X_L = X_C \Rightarrow \cos \phi = 0$$

Impedance $\rightarrow$ purely Reactive

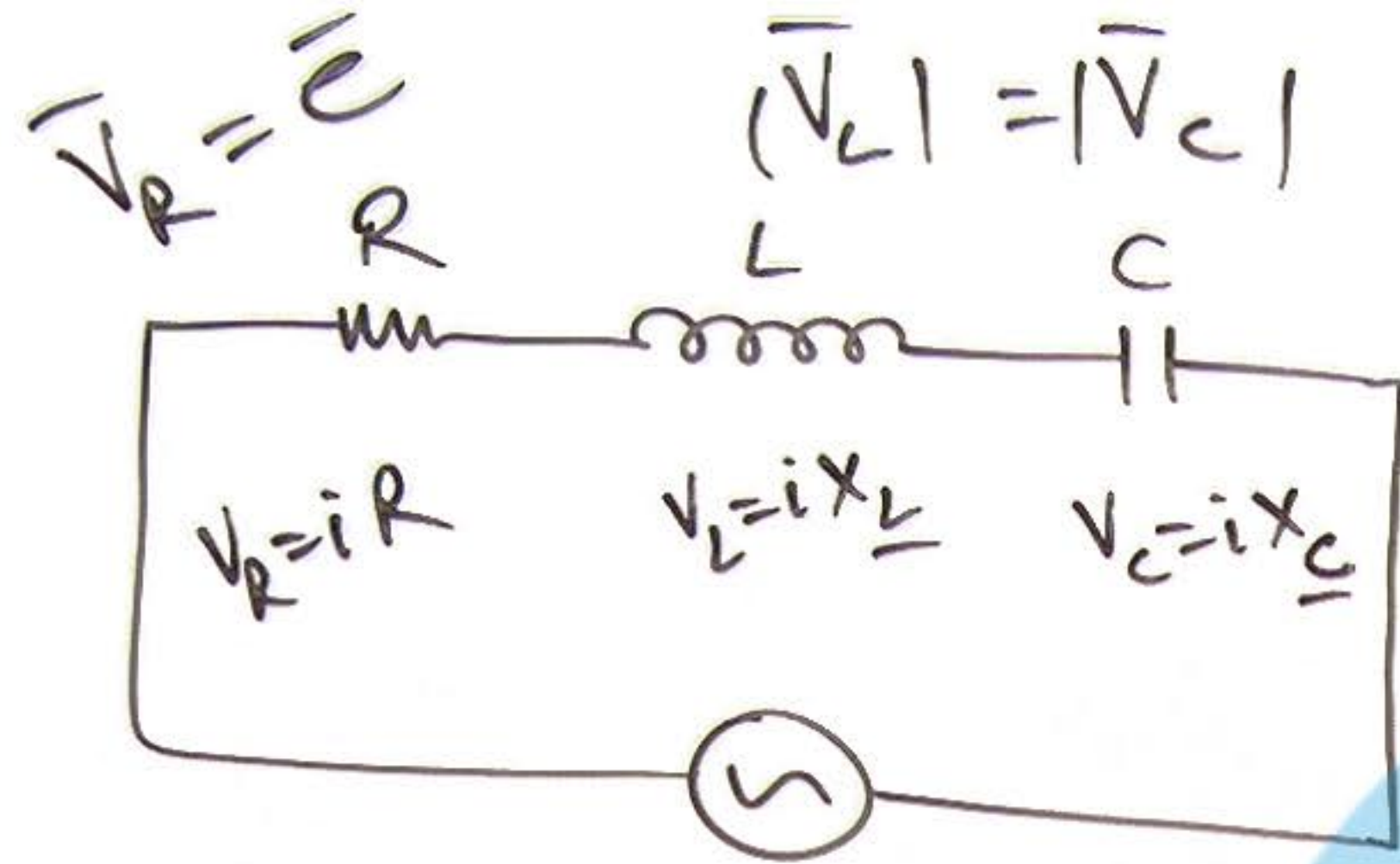
$$\Rightarrow \underline{P_{avg} = 0} [\text{Min}]$$

$$\cos \phi = \left| \frac{R}{\sqrt{R^2 + X^2}} \right|$$

$$\tan \phi = \left| \frac{X}{R} \right|$$



13 Power in Series RLC circuit:



$$e = e_0 \sin \omega t$$

$$Z = \sqrt{R^2 + \left(\omega L - \frac{1}{\omega C}\right)^2}$$

$$i_0 = \frac{e_0}{Z} \quad \& \quad i_{rms} = \frac{e_{rms}}{Z}$$

$$P_{avg} = \text{Cons. loss} \cdot \frac{R}{\sqrt{R^2 + \left(\omega L - \frac{1}{\omega C}\right)^2}}$$

P_{avg} will be max when $\omega L = \frac{1}{\omega C}$

Resonance
in RLC Series
Ckt

Resonant
frequency

$$\Rightarrow \omega_R = \frac{1}{\sqrt{LC}} \quad \checkmark$$

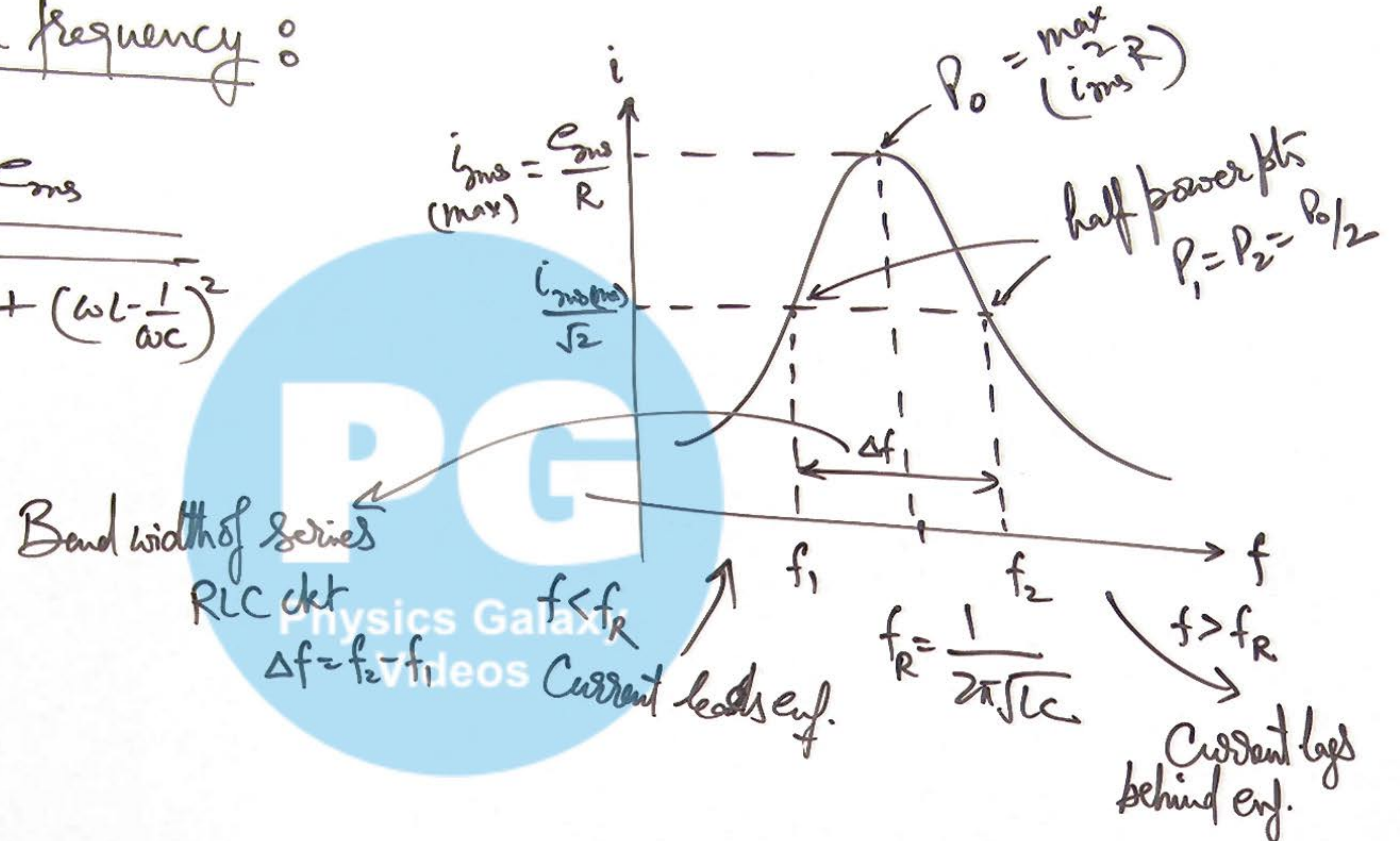
$$\Rightarrow f_R = \frac{1}{2\pi\sqrt{LC}}$$

When $Z = R \Rightarrow P_{avg} = \underline{\text{max.}}$

⑭ Current variation in Series RLC ckt

with frequency :

$$i_{rms} = \frac{E_{rms}}{\sqrt{R^2 + \left(\omega L - \frac{1}{\omega C}\right)^2}}$$



⑮ Band width & Quality factor of series RLC ckt :

Band width.

$$\Delta\omega = \omega_2 - \omega_1 = \frac{R}{L}$$

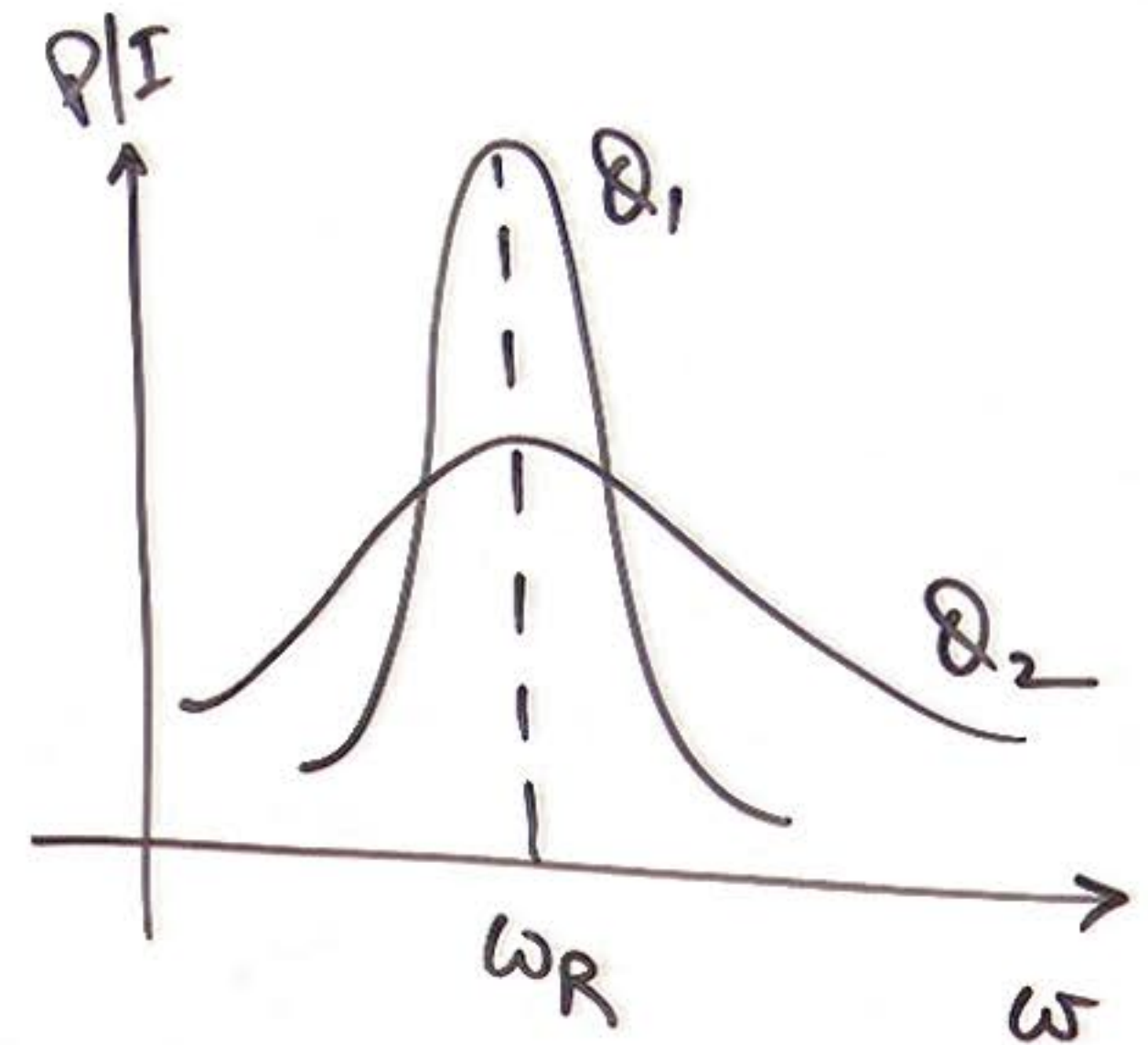
Half power freq.

Quality factor
(Q-factor)

$$Q = \frac{\omega_R}{\Delta\omega} = \frac{1}{\sqrt{LC}} \cdot \frac{L}{R} = \frac{1}{R} \sqrt{\frac{L}{C}}$$

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⑯ Selectivity of Series RLC ckt.



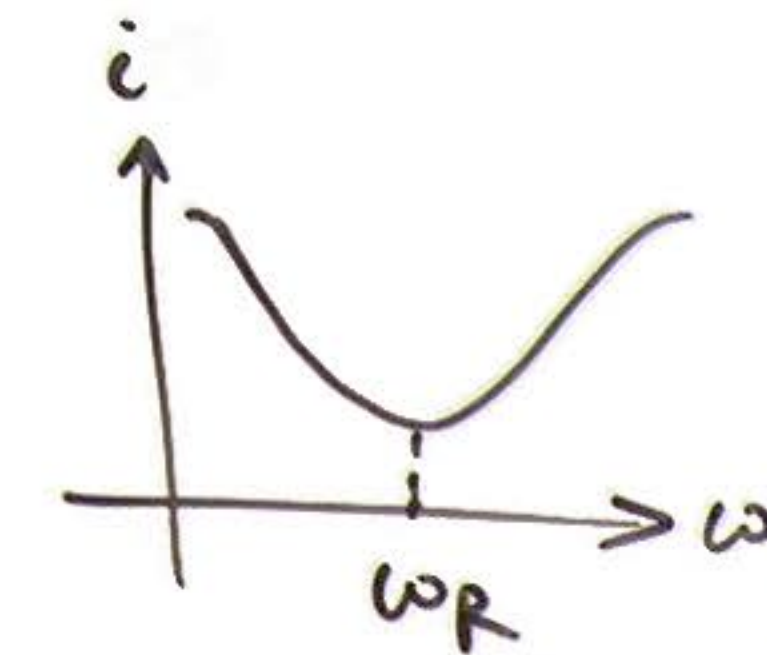
$Q_1 > Q_2$
(Selectivity)₁ > (Selectivity)₂

16 Parallel RLC circuit:

$$i_R = \frac{e}{R}; i_L = \frac{e}{X_L}; i_C = \frac{e}{X_C}$$

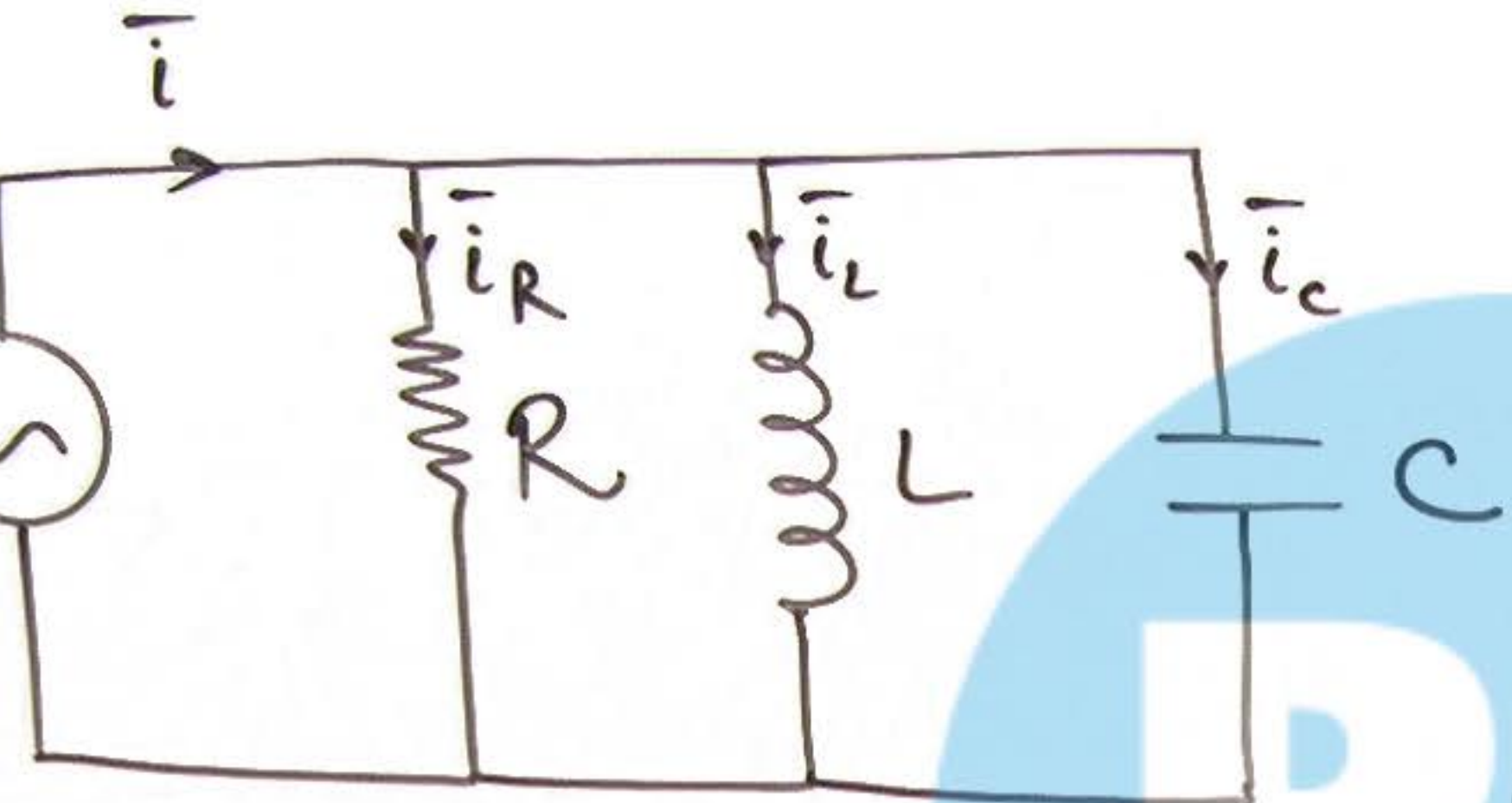
$$\frac{1}{Z} = \frac{1}{R \angle 0^\circ} + \frac{1}{\omega L \angle 90^\circ} + \frac{1}{\frac{1}{\omega C} \angle -90^\circ}$$

$$\Rightarrow \frac{1}{Z} = \frac{1}{R} \angle 0^\circ + \frac{1}{\omega L} \angle -90^\circ + \frac{1}{\frac{1}{\omega C}} \angle 90^\circ$$



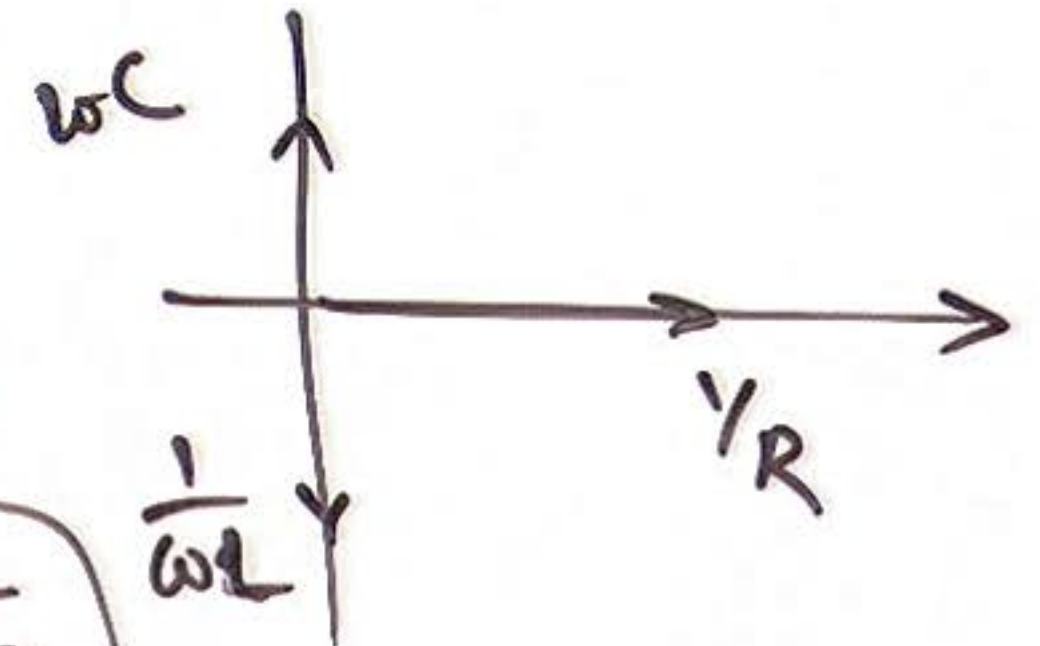
$$e = e_0 \sin \omega t$$

(Rejector ckt)



$$\frac{1}{Z} = \sqrt{\left(\frac{1}{R}\right)^2 + \left(\omega C - \frac{1}{\omega L}\right)^2}$$

$$i = e \sqrt{\left(\frac{1}{R}\right)^2 + \left(\omega C - \frac{1}{\omega L}\right)^2}$$



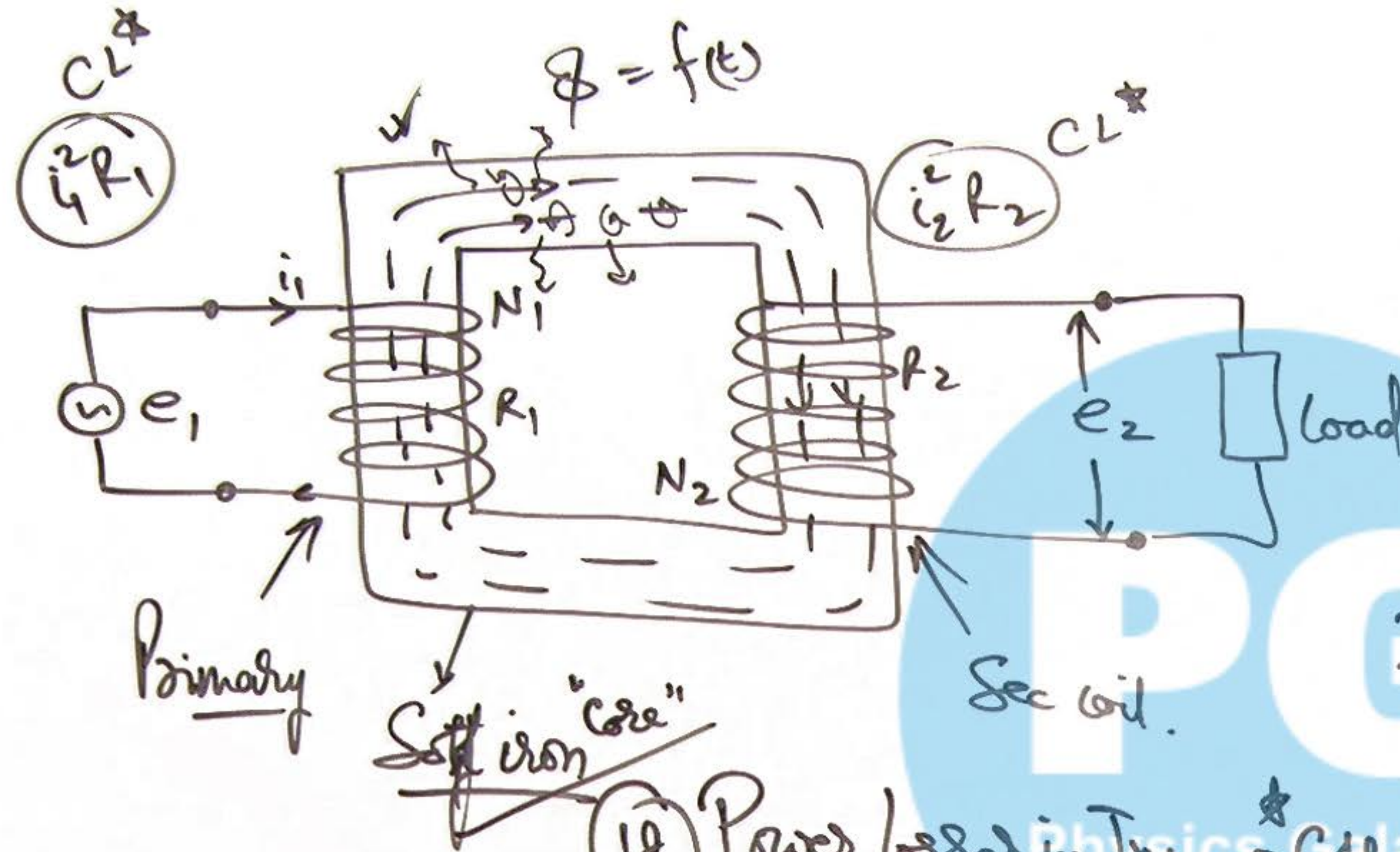
$$\bar{i} = \bar{i}_R + \bar{i}_L + \bar{i}_C$$

$$i = \sqrt{i_R^2 + (i_L - i_C)^2}$$

At Resonance

$$\omega_R = \frac{1}{\sqrt{LC}} \rightarrow i_{\min.}$$

(17) Transformer:



Transformer eqⁿ

$$\frac{e_1}{e_2} = \frac{N_1}{N_2}$$

if $e_2 < e_1 \rightarrow$ Step down T_x

if $e_2 > e_1 \rightarrow$ Step up T_x

Ideal transformer (No power losses)

$$i_1 e_1 = i_2 e_2$$

(18) Power losses in T_x : → Copper loss

→ Iron loss (eddy current loss)